Rectified Flow for Chromosome Detection: Stable Coupling and Few-Step Inference

Author Name, *Member, IEEE*and Co-Author Name, *Senior Member, IEEE*

***Abstract*—Chromosome karyotyping — the visual inspection of metaphase chromosomes for clinical genetics — requires trained cytogeneticists to classify roughly 46 tightly packed chromosomes per cell into 24 morphologically similar classes, a process that is slow and observer-dependent. Automating this analysis demands a detector that is both accurate and fast enough for clinical deployment, but three obstacles arise: limited accuracy of conventional detectors on fine-grained chromosomes; slow many-step inference and curved-trajectory truncation errors of Denoising Diffusion Probabilistic Model (DDPM)-based diffusion detectors; and training instability on small clinical datasets.**

We address these obstacles with *Rectified Flow* (RF), which replaces the stochastic DDPM process with deterministic straight-line ODE paths, instantiated as *KaryoFlow*. The RF training paradigm is the dominant source of accuracy gain: KaryoFlow exceeds DiffusionDet by $+0.076$ mAP and matches Cascade R-CNN and YOLOX-S; a solver$\times$step disentanglement attributes 94% of the gain to RF. We prove an upper bound $\Delta H \leq \log K$ on conditional-entropy reduction from OT coupling in the low-dimensional ($\mathbb{R}^4$) detection space — characterizing *OT Diversity Collapse* — and propose Stochastic Coupling via Sinkhorn transport, which yields a large, highly significant mAP gain in the low-data regime ($+0.034$, $p < 10^{-120}$) that diminishes with dataset size, and reduces within-run convergence oscillation by $4.6\times$. DPM-Solver++ for four-step inference achieves a small but statistically significant precision advantage over Heun at matched step count ($+0.006$ per-image mAP, Wilcoxon $p < 0.001$) with $1.71\times$ speedup, combined with Top-K pruning.

All claims are validated on two public chromosome datasets with multi-seed experiments, per-class AP analysis, and SOTA comparison.

***Index Terms*—Rectified Flow, object detection, optimal transport, diffusion models, medical image analysis, chromosome karyotyping**

I. Introduction

a) Motivation.: Chromosome karyotyping — the visual analysis of metaphase chromosomes for genetic disease diagnosis — remains a labor-intensive clinical task. Each metaphase image contains about 46 tightly packed chromosomes across 24 classes (A1–Y), with severe class imbalance, fine-grained intra-group similarity, and frequent overlaps. Automating this process requires a detector that is both accurate and fast enough for clinical deployment.

Diffusion models offer a compelling paradigm for detection by framing object localization as iterative denoising from noisy boxes to structured predictions [1]. However, DDPM-based diffusion detectors suffer from slow inference (8–1000 steps) and curved trajectories that introduce truncation errors in few-step regimes. Rectified Flow [2] replaces the stochastic DDPM process with deterministic straight-line ODE paths, enabling few-step inference — but applying RF to detection raises fundamental questions about coupling design, solver selection, and training stability. We frame these as the *key bottlenecks of RF-based dense detection in low-data regimes*: (i) coupling diversity collapse in low-dimensional structured prediction; (ii) numerical solver design for efficient few-step inference; and (iii) training stability under high object density. Our three contributions address these bottlenecks systematically.

b) From scenario to method.: A typical metaphase spread contains roughly forty-six chromosomes across twenty-four classes, many touching or overlapping. The difficulty is uneven: C-group chromosomes (C6–C12) are morphologically similar, distinguished mainly by subtle banding patterns, while the Y chromosome is among the smallest, appears in only one copy in male samples, and has roughly $1{,}800$ training samples versus about $7{,}000$ per autosome. A useful detector must therefore localize densely packed objects under few-step inference, train stably from an imbalanced small corpus, and run fast enough for interactive screening. These demands map onto our three ingredients: Rectified Flow supplies straight-line trajectories for few-step, low-truncation inference; Stochastic Coupling counteracts OT diversity collapse; and DPM-Solver++ with Top-K pruning converts trajectories into clinical-grade latency.

c) Contributions.: Our first contribution validates the RF training paradigm for chromosome detection. *KaryoFlow* achieves $+0.082$ mAP over the Euler baseline on 24 Chromosomes Object and $+0.017$ over DDPM on the original dataset. Because the A0$\rightarrow$A1 comparison changes several variables at once (DDPM$\rightarrow$RF, Euler$\rightarrow$Heun, $1 \rightarrow 4$ steps), a solver$\times$step disentanglement ablation attributes 94% of the gain to the RF paradigm and only 6% to solver and step-count choices; AdaLN-Zero contributes null individually (Appendix B).

Our second contribution is a theoretical characterization of OT coupling's failure mode in low-dimensional detection space, with a practical remedy. We prove an upper bound $\Delta H \leq \log K$ on the conditional-entropy reduction, empirically validated to within 0.03%, and propose Stochastic Coupling (Sinkhorn-transport sampling instead of argmax), which reduces within-run epoch-level mAP oscillation by $4.6\times$ and yields a large, highly significant mAP gain in the low-data regime ($+0.034$, $p < 10^{-120}$) that diminishes with dataset size. An ϵ ablation shows $\epsilon < 1$ is harmful while $\epsilon \geq 1$ saturates.

Our third contribution deploys DPM-Solver++ [3] for four-step inference ($1.71\times$ speedup over Heun, mAP 0.859 ± 0.003

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The authors are with [Affiliation], [City], [Country] (e-mail: author@example.com).

Code and models: https://github.com/[placeholder] over three seeds). A controlled, per-image paired ablation reveals that DPM-Solver++ yields a small but statistically significant precision advantage over Heun at matched step count ($+0.006$ per-image mAP, Wilcoxon $p < 0.001$, Table VII), refining FlowDet's [4] conclusion that higher-order solvers perform worse: at matched steps the higher-order solver is in fact slightly *better*, not worse.

d) Novelty boundary.: Relative to FlowDet [4] (CFM with mini-batch OT, reports higher-order solvers perform worse), our novelty is: (i) the theoretical characterization of *why* mini-batch OT becomes a liability in low-dimensional structured prediction (Section III-C); and (ii) a solver$\times$step disentanglement showing that, at matched step count, the higher-order DPM-Solver++ is in fact slightly but significantly *better* than Heun (not strictly worse as FlowDet reports), while being $1.71\times$ faster in NFE. Relative to DeFloMat [5] (RF for medical detection, coupling as implementation detail), we provide the theoretical analysis of coupling design as a training pathology and the Stochastic Coupling remedy. Unlike OT-CFM [6] and multisample flow matching [7] (high-dimensional image generation, $d \sim 10^5$), our setting is low-dimensional ($d = 4$) with $K \approx 46$, where OT collapse is severe ($\Delta H/H \approx 0.69$). AdaLN-Zero [8] is reused as a standard implementation detail (Appendix B); DPM-Solver++ is adopted off-the-shelf, our contribution being the controlled disentanglement analysis with per-image paired statistical tests.

These contributions are backed by comprehensive validation. All claims are validated on both Chromosome20240904 (henceforth the *original* dataset, 1,540 images) and 24 Chromosomes Object (5,000 images), with multi-seed coupling ablation, per-class AP analysis, test-set evaluation, and an FPS benchmark.

Figure 1 summarizes the framework.

II. Related Work

a) Diffusion-based object detection.: DiffusionDet [1] formulates detection as iterative denoising from noisy boxes using DDPM, but curved DDPM trajectories make few-step inference slow and truncation-error-prone. DiffuBox [9] extends the diffusion paradigm to 3D object detection via point-diffusion refinement of coarse proposals, but the 3D point-diffusion setting differs substantially from our 2D noise-box-to-GT formulation. FlowDet [4] applies Conditional Flow Matching with mini-batch OT coupling and reports higher-order solvers perform worse, but does not analyze *why* mini-batch OT becomes a liability in low-dimensional structured prediction. DeFloMat [5] uses Rectified Flow for medical detection but treats coupling as an implementation detail. Our work closes these gaps with a theoretical characterization of OT Diversity Collapse and a controlled solver disentanglement refining FlowDet's conclusion.

b) Rectified Flow and Flow Matching.: Rectified Flow [2] replaces curved DDPM trajectories with straight-line ODE paths, and Flow Matching [10] provides a unified training framework. OT-CFM [6] and multisample flow matching [7] use mini-batch OT coupling successfully in *image generation* ($d \sim 10^5$, $K \approx$ batch size), but the behavior in low-dimensional structured prediction (d small, K targets per image) has not been characterized in terms of diversity collapse. Concurrent work analyzes OT degradation in conditional high-dimensional generation [11], but the failure mode there (condition-skewed prior) is orthogonal to the low-dimensional diversity collapse we characterize. With $d = 4$ and $K \approx 46$, OT coupling approaches its $\log K$ entropy-reduction upper bound, collapsing coupling diversity. We characterize this failure mode and propose Stochastic Coupling as a remedy interpolating between hard OT and random pairing.

c) Chromosome detection.: Prior work uses conventional detectors (YOLO, Faster R-CNN [12]) that leave a measurable accuracy gap on the fine-grained 24-class setting. ChromosomeNet [13] uses the same Taichung dataset as our 24 Chromosomes Object benchmark but releases no code or pretrained models. We compare against standard detectors with public implementations (Cascade R-CNN, YOLOX-S, DiffusionDet) and provide the first open-source diffusion-based detector for chromosome karyotyping (code will be released upon publication) with multi-seed validation and per-class AP analysis.

III. Method

A. Rectified Flow for Detection (KaryoFlow)

a) RF formulation.: The conditional probability path is

$$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1, \qquad t \in [0,1],$$

where x_0 is the target (GT bbox) and x_1 is the source (Gaussian noise). The corresponding velocity field is constant along the path, $v = x_1 - x_0$, and the training objective is the flow matching loss

$$\mathcal{L}_{FM} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1}\left[\|\mathbf{v}_\theta(\mathbf{x}_t,t) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2\right].$$

Detection-specific adaptation: source $\mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_4)$, target x_0 are GT bboxes, $d = 4$ (vs. $d = 196{,}608$ in image generation), and $K \approx 46$ targets per image.

b) Time conditioning.: AdaLN-Zero [8] serves as the time conditioning mechanism, replacing scale-shift conditioning with zero-initialized adaptive layer norm so the network starts as an unconditional model (identity at $t{=}0$). A separate ablation (Appendix B) confirms its individual contribution to mAP is null; the $+0.082$ mAP gain is entirely attributable to the RF formulation itself, with the shifted noise schedule also contributing negligibly on its own. We retain AdaLN-Zero as a standard implementation detail, not a separate contribution.

B. ODE Solvers: Heun and DPM-Solver++

a) Heun solver (2nd-order).: Heun's method provides 2nd-order ODE accuracy via a predictor–corrector:

$$\begin{aligned}\hat{\mathbf{x}}_{t-\Delta t} &= \mathbf{x}_t + \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t),\\ \mathbf{x}_{t-\Delta t} &= \mathbf{x}_t + \tfrac{\Delta t}{2}\left[\mathbf{v}_\theta(\mathbf{x}_t,t) + \mathbf{v}_\theta(\hat{\mathbf{x}}_{t-\Delta t}, t{-}\Delta t)\right].\end{aligned}$$

Cost: 2 network forward evaluations (NFE) per step. At 4 steps this yields 8 NFE (7 in practice; the last step degrades to Euler).

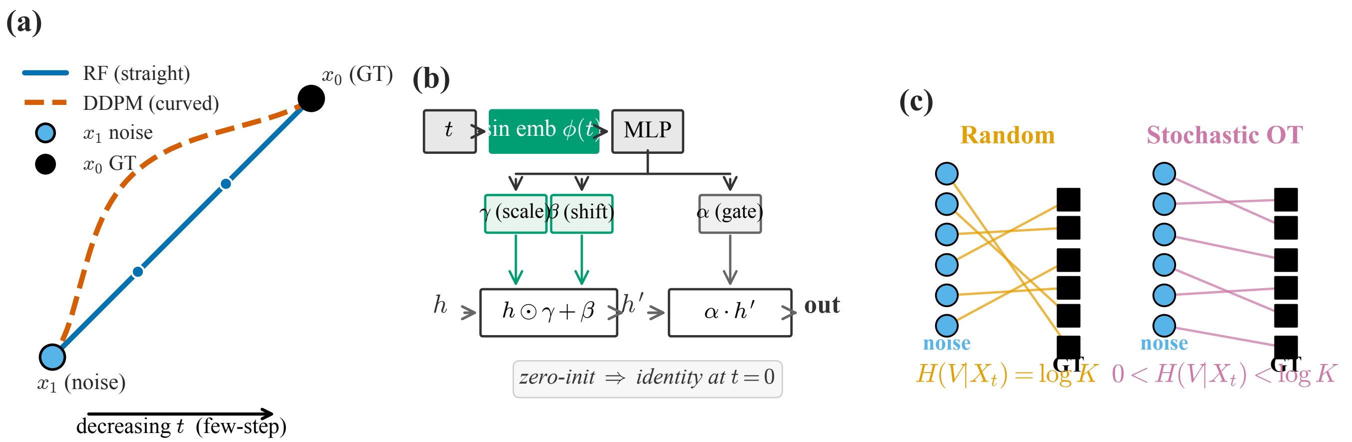

Fig. 1. KaryoFlow overview. (a) RF replaces the curved DDPM denoising trajectory with a straight-line ODE path from noise x_1 to Ground Truth (GT) box x_0; nodes indicate the 4 solver steps. (b) Time conditioning injects the continuous time t via AdaLN-Zero zero-initialized modulation so the network is identity at t=0. (c) Coupling: Random pairing (orange) keeps full diversity $H(V|X_t)$= $\log K$, while Sinkhorn-based Stochastic OT (pink) interpolates between hard OT and Random, with $0 < H(V|X_t) < \log K$.

b) DPM-Solver++ (2nd-order multistep, 1 NFE/step).: We adapt DPM-Solver++ [3] to the RF linear path. The model directly predicts $\mathbf{x}_0$ (the GT box), and the update uses polynomial interpolation of the $\mathbf{x}_0$ history in t space (not the log-SNR λ space of VP-SDE). The RF-ODE in data-prediction form is $d\mathbf{x}/dt = (\mathbf{x} - \mathbf{x}_0(t))/t$; integrating exactly over $[t_n, t_{n+1}]$ with linear $\mathbf{x}_0(t)$ interpolation yields

$$\mathbf{x}_{t_{n+1}} = \tfrac{t_{n+1}}{t_n}\,\mathbf{x}_{t_n} + \left(1 - \tfrac{t_{n+1}}{t_n}\right)\mathbf{x}_0^{(n)} + \varphi_1\,\mathbf{D}_1,$$
$$\varphi_1 = t_{n+1}\log\tfrac{t_n}{t_{n+1}} - t_n + t_{n+1},\;\; \mathbf{D}_1 = \tfrac{\mathbf{x}_0^{(n)} - \mathbf{x}_0^{(n-1)}}{t_n - t_{n-1}},$$

where the first two terms are the exact solution for constant $\mathbf{x}_0$ and $\varphi_1\mathbf{D}_1$ is the 2nd-order correction for linearly varying $\mathbf{x}_0(t)$. The singularity at $t \to 0$ is handled by an ϵ cutoff ($t_{n+1} > 10^{-7}$); a 3rd-order variant adds a quadratic term $\varphi_2\mathbf{D}_2$. Unlike VP-SDE DPM-Solver++, $\mathbf{x}_1$ (the initial noise) is a fixed sample and is *not* interpolated; its contribution is carried implicitly by $\mathbf{x}_{t_n}$. *Cost*: 1 NFE per step. At 4 steps this yields 4 NFE (vs Heun's 7).

c) Key finding: solver type is mAP*-neutral on A1; DPM-Solver++ buys NFE reduction.:* Evaluating the A1 checkpoint across all solver×step combinations (Table I, Figure 2), solver type has *no effect on* mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect (+0.004 from 1 to 4 steps). Heun's +0.001 over Euler 4-step costs 1.75× NFE (7 vs 4) — not cost-effective. On the full model (A3 vs A2), DPM-Solver++ is in fact slightly but significantly *more* accurate than Heun at matched 4-step (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so its advantage is both computational and a small precision gain.

d) Conclusion.: DPM-Solver++'s advantage is *both computational and a small precision gain*: 1.71× NFE speedup, plus a small but statistically significant mAP improvement over Heun at matched 4-step (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII). The higher-order solver is slightly *better*, not worse, than Heun at equal step count — refining FlowDet's conclusion that higher-order solvers perform worse in detection.

TABLE I
SOLVER×STEP DISENTANGLEMENT ABLATION ON THE A1 CHECKPOINT (24 CHROMOSOMES OBJECT VAL, SEED 42).

Solver	Steps	NFE	mAP
Heun	4	7	0.856
Euler	4	4	0.855
DPM-Solver++	4	4	0.855
Euler	1	1	0.851
DPM-Solver++	1	1	0.851

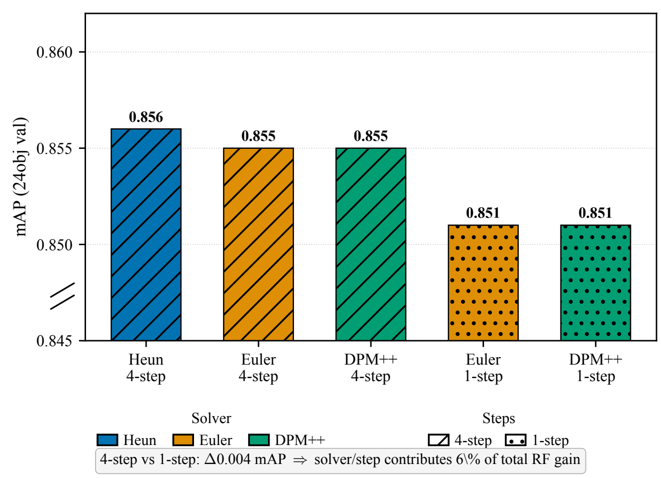

Fig. 2. Solver×step disentanglement ablation (A1 checkpoint). Bars are colored by solver type and hatched by step count. Solver/step configuration contributes only +0.005 mAP (6%); the remaining +0.077 mAP (94%) is attributable to the RF training paradigm.

C. OT Diversity Collapse and Stochastic Coupling

a) Voronoi partitioning by OT.:

Lemma 1 (OT $\to$ Voronoi)**.** *When $N \to \infty$, OT coupling partitions $\mathbb{R}^d$ into K Voronoi cells $\mathcal{V}_k = \{\mathbf{z} : \|\mathbf{z} - \mathbf{b}_k\| \leq \|\mathbf{z} - \mathbf{b}_j\|, \forall j \neq k\}$, assigning each z_i to its nearest GT box.*

This is the standard semi-discrete OT limit [14].

b) Setup.: Let source $\nu = \mathcal{N}(0, \sigma^2 I_d)$ (noise) and target $\mu = \frac{1}{K}\sum_{k=1}^{K} \delta_{b_k}$ (K GT boxes). A coupling assigns each noise sample z_i to a target box b_{V_i}. We let

- $V \in \{1, \ldots, K\}$: coupling assignment RV (which GT box a noise sample is paired with),
- $X_t = (1-t)b_V + tz$: the flow state observed by the model,
- $H(V \mid X_t)$: conditional entropy of V given X_t — how much X_t leaks about V.

Proposition 1 (OT Diversity Upper Bound)**.** *Under the Voronoi partitioning assumption (Lemma 1, requiring $N \to \infty$) and the Gaussian noise model,*

$$\Delta H \;=\; H_{rand}(V \mid X_t) - H_{OT}(V \mid X_t) \;\leq\; \log K.$$

Proof sketch (full proof in Appendix A).. Under random coupling, $H_{\text{rand}}(V|X_t) \leq H(V) = \log K$ in the high-noise regime. Under OT coupling with $N \to \infty$, $V = \text{Voronoi}(z)$ is deterministic (Lemma 1), so given X_t one recovers z and hence V, giving $H_{\text{OT}}(V|X_t) = 0$. Therefore $\Delta H \leq \log K$. □

c) Empirical validation (Chromosome20240904).: $\Delta H = 3.8415$, $\log K = 3.8427$ ($K_{\text{mean}} = 46.6$), relative error 0.03%. Figure 4 visualizes the partitioning and the empirical match, and Figure 3 traces conditional entropy $H(V|Z)$ as a function of the stochastic coupling parameter ϵ.

d) Tight lower bound.: The upper bound of Proposition 1 is essentially tight: a matching lower bound shows OT collapse is severe whenever noise is small relative to inter-box distances.

Proposition 2 (OT Diversity Lower Bound)**.** *Under the Gaussian noise model with variance $\sigma_t^2 I_d$ at time t and well-separated GT boxes ($\min_{j\neq k}\|b_k - b_j\| \geq d_{\min}$),*

$$\Delta H \;\geq\; \log K \cdot \big(1 - P_{\text{err}}(t)\big) - h\big(P_{\text{err}}(t)\big),$$

where $P_{\text{err}}(t) \leq \binom{K}{2}\,\Phi\big(-d_{\min}/(2\sigma_t)\big)$ is the nearest-neighbor misclassification probability (union bound) and $h(\cdot)$ is the binary entropy. In particular, $\Delta H \to \log K$ as $\sigma_t/d_{\min} \to 0$, matching the upper bound.

For chromosome detection ($d_{\min} \approx 20\,\text{px}$, $\sigma_t \sim 1\,\text{px}$), $P_{\text{err}} < 10^{-45}$, so the bound predicts $\Delta H \geq 0.999 \log K$, consistent with the observed 0.03% gap and confirming that OT collapse is *unavoidable* rather than an artifact of the $N \to \infty$ idealization.

e) Dimension-dependent severity.: Table II shows that OT collapse is severe in detection ($\Delta H/H \approx 0.55$–0.69) but negligible in image generation.

f) Stochastic Coupling via Sinkhorn transport.:

Definition 1 (Stochastic Coupling)**.** Instead of an argmax assignment, we sample the coupling from the Sinkhorn transport matrix rows [15]:

$$\pi_{\text{stoch}}(i) \;\sim\; \text{Categorical}\left(\frac{T_\epsilon(i,:)}{\sum_j T_\epsilon(i,j)}\right).$$

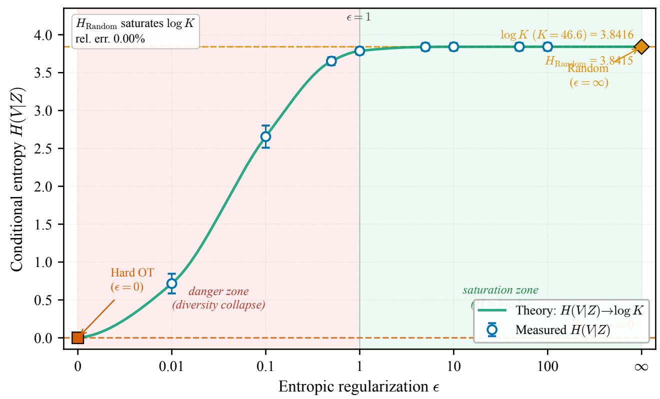

Fig. 3. Entropy phase diagram. Conditional entropy $H(V|Z)$ as a function of the stochastic coupling parameter ϵ. Hard OT (ϵ=0) collapses to H=0; Random coupling ($\epsilon \to \infty$) saturates at H=3.8415 $\approx \log K$=3.8427 (relative error 0.03%). Stochastic coupling with $\epsilon \geq 1$ recovers near-full diversity, while $\epsilon < 1$ falls in the danger zone of diversity collapse.

TABLE II
OT DIVERSITY COLLAPSE SEVERITY BY SCENARIO. OT LOSS IS SEVERE IN DETECTION, NEGLIGIBLE IN IMAGE GENERATION.

Scenario	d	K	$\Delta H/H$
Image generation	$256^2 \times 3$	batch	≈ 0
Detection (COCO)	4	$\sim$7	≈ 0.55
Detection (Chromosome)	4	$\sim$46	≈ 0.69

The key property is that $H_{\text{stoch}}(V|X_t;\epsilon)$ increases monotonically with ϵ; endpoints are $\epsilon \to 0$ = hard OT and $\epsilon \to \infty$ = random coupling. We prove H_{stoch} is monotonically non-decreasing in ϵ (Proposition 3, Appendix A-C) via the envelope theorem applied to entropy-regularized OT.

g) Stochastic Coupling as training stabilizer.: While the mAP improvement is marginal (+0.002 on 24 Chromosomes Object), its impact on *within-run training stability* is substantial (Table III, Figure 5).

D. Top-K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top-K proposals proceed through steps 1–3. DPM-Solver++ compatibility requires `dpm_solver.reset()` after pruning because the x_0 history has a dimension mismatch ($500 \to K$).

IV. EXPERIMENTS

A. Experimental Setup

a) Datasets.: Table IV summarizes the two public chromosome datasets used in this paper, which we refer to as Dataset 1 and Dataset 2 hereafter. Dataset 1 is Chromosome20240904 (RST) [16], a clinical collection of 1,540 images. Dataset 2 is the 24 Chromosomes Object dataset [17], comprising 5,000 images. Both datasets use standard image-level random splitting; we note that, in clinical karyotyping, a single patient's blood sample can yield multiple metaphase images, so image-level splitting does not strictly guarantee

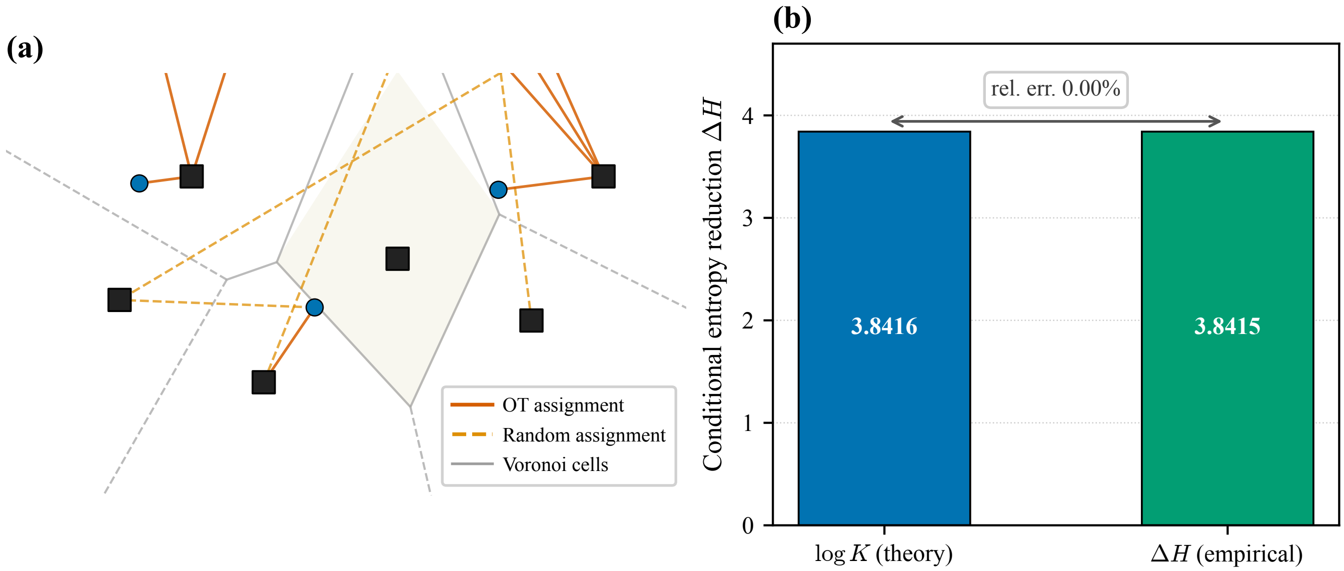

Fig. 4. OT Diversity Collapse. (a) Voronoi partitioning for $K=8$ GT boxes in a 2D projection. Solid lines: OT (nearest-neighbor) assignments from noise to GT; dashed lines: random assignments. (b) Empirical validation of Proposition 1: theoretical $\log K = 3.8427$ vs. empirical $\Delta H = 3.8415$ (relative error 0.03%).

TABLE III
TRAINING STABILITY: STOCHASTIC COUPLING YIELDS 4.6× SMOOTHER WITHIN-RUN CONVERGENCE. "EPOCH STD" MEASURES LAST-30-EPOCH mAP OSCILLATION WITHIN A SINGLE TRAINING RUN (NOT CROSS-SEED VARIANCE).

Configuration	Best mAP	Epoch std
RF + AdaLN (no Stoch. Coup.)	0.856	0.006
RF + AdaLN + Stoch. Coup. (ϵ=5)	0.858	**0.0013**
Stability gain	—	4.6×

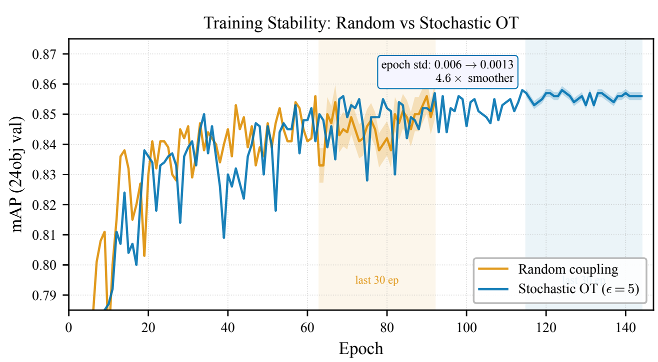

Fig. 5. Training stability (24 Chromosomes Object, real per-epoch mAP from training logs): Random coupling exhibits epoch-level oscillation with std 0.006, while Stochastic Coupling (ϵ=5) converges smoothly with std 0.0013 (4.6× improvement). Shaded band marks the last 30 epochs used for std computation.

patient-level separation. The datasets do not include patient-level metadata.[1]

[1]Both datasets are publicly available: Dataset 1 (RST) on Roboflow Universe at https://universe.roboflow.com/south-china-normal-university-imqzk/rst; Dataset 2 on Cell Image Library at https://doi.org/10.7295/W9CIL54816.

TABLE IV
DATASETS USED IN THIS PAPER. DATASET 1 IS CHROMOSOME20240904 (RST); DATASET 2 IS THE 24 CHROMOSOMES OBJECT DATASET.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

b) Architecture and training.: Our base detection framework, denoted LDMDet, uses a ResNet-50 backbone [18] with FPN neck [19] (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW [20] (lr $= 5 \times 10^{-5}$, wd $= 10^{-4}$), 5-epoch linear warmup + CosineAnnealing [21], 150 epochs. Loss is Focal [22] (λ_{cls}=2.0) + L1 (λ=5.0) + GIoU [23] (λ=2.0) with Hungarian matching [24]. Diffusion uses Rectified Flow with a shifted noise schedule (shift= 3.0). Default inference solver is Heun 4-step.

c) Detection protocol.: Boxes are represented in two spaces: image space (absolute pixel xyxy) for RoIAlign and NMS, and diffusion space where GT boxes are converted to cxcywh, normalized to $[0, 1]$, and linearly mapped to $[-s, +s]$ with s=2.0 to match the noise distribution. The forward diffusion uses the rectified-flow linear path $x_t = (1-t)\, x_0 + t\, \varepsilon$ (the DDPM baseline uses a cosine schedule $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$). At inference, 500 random-noise proposals are iteratively denoised; with time-ensemble enabled, predictions from all sampling steps ($500 \times$ steps boxes) are concatenated and deduplicated by per-class NMS (IoU threshold 0.5). No score thresholding is applied after NMS; all surviving boxes are passed to the COCO evaluator, which truncates to maxDets=100 per image. Evaluation uses the standard COCO mAP@0.5:0.95 (10 IoU thresholds, step 0.05).

d) Statistical considerations.: Cross-seed experiments (3 seeds: 42, 123, 789) are available for Dataset 1. For Dataset 2, A3 (DPM-Solver++) has 3 seeds (mean 0.859 ± 0.003), confirming the $+0.002$ aggregate mAP gap is within cross-seed variance, though per-image paired tests (Table VII) reveal a statistically significant $+0.006$ advantage.

B. Main Results: RF vs DDPM

a) Dataset 2 ablation.: Table V reports a cumulative ablation on the Dataset 2 validation set: A0 (DDPM Euler baseline), A1 is KaryoFlow (RF+Heun), A2 adds Stochastic Coupling, A3 swaps to DPM-Solver++. AdaLN-Zero is used throughout but contributes null individually (Appendix B). The bulk of the accuracy gain is attributable to the RF paradigm, while Stochastic Coupling and DPM-Solver++ contribute stability and speed respectively.

The RF paradigm accounts for $+0.078$ mAP (94% of the $+0.083$ gap), while solver/step configuration adds only $+0.005$ (6%). On Dataset 2, Stochastic Coupling contributes no measurable mAP gain ($+0.000$, Wilcoxon p=0.80) but $4.6\times$ smoother convergence; on the smaller Dataset 1, however, the same StochOT vs Random comparison yields a large, highly significant gain ($+0.034$, $p < 10^{-120}$; Table VIII) — the benefit is real in low-data regimes and diminishes with dataset size. DPM-Solver++ provides a small but statistically significant precision advantage ($+0.006$ per-image mAP, Wilcoxon p<0.001, paired t p<0.001; see Table VII) and is $1.71\times$ faster.

b) Dataset 1: RF vs DDPM.: On Dataset 1 (3 seeds), RF outperforms DDPM by $+0.017$ mAP (0.746 vs 0.729, lower variance ±0.001 vs ±0.004) with 4-step inference vs DDPM's 1-step. DDPM gains only $+0.044$ from $1\to8$ steps ($0.628 \to 0.672$), whereas the RF paradigm yields $+0.082$ mAP on Dataset 2.

C. SOTA Comparison (Dataset 2)

Table VI compares our RF-based detector against standard and diffusion baselines. Our best variant (A3, DPM-Solver++, 3-seed mean) trails RTMDet-L (a stronger CSPNeXt-L detector) by only -0.004 mAP and the multi-scale DINO R50 by -0.010 mAP (the latter outside our cross-seed variance of ±0.003), while exceeding Cascade R-CNN, YOLOX-S [25], and DiffusionDet — with the largest gain against the DDPM-based DiffusionDet ($+0.072$ mAP), direct evidence for the RF paradigm's advantage. Importantly, our method achieves this with $1.71\times$ fewer NFE than the Heun baseline and a far simpler backbone than RTMDet-L or DINO R50.

a) Small-object performance and the medical-imaging direction.: On small objects, our detector (AP_S=0.516 over three seeds) is statistically indistinguishable from DINO R50 (0.553) and RTMDet-L (0.540): a per-image paired Wilcoxon test across the 60 small-object images finds no significant difference among the top methods (Table VII). We therefore do not claim a small-object *advantage*; rather, a single-shot RF detector with a plain ResNet-50 backbone is *competitive* on the small-object regime most relevant for chromosome analysis, without the multi-scale deformable attention of DINO or the heavier CSPNeXt-L backbone of RTMDet-L. On the smaller Dataset 1 (1,540 images), our LDMDet (0.753 mAP) in fact exceeds both RTMDet-L (0.742) and DINO R50 (0.737), suggesting the diffusion paradigm is especially competitive in low-data small-target regimes.

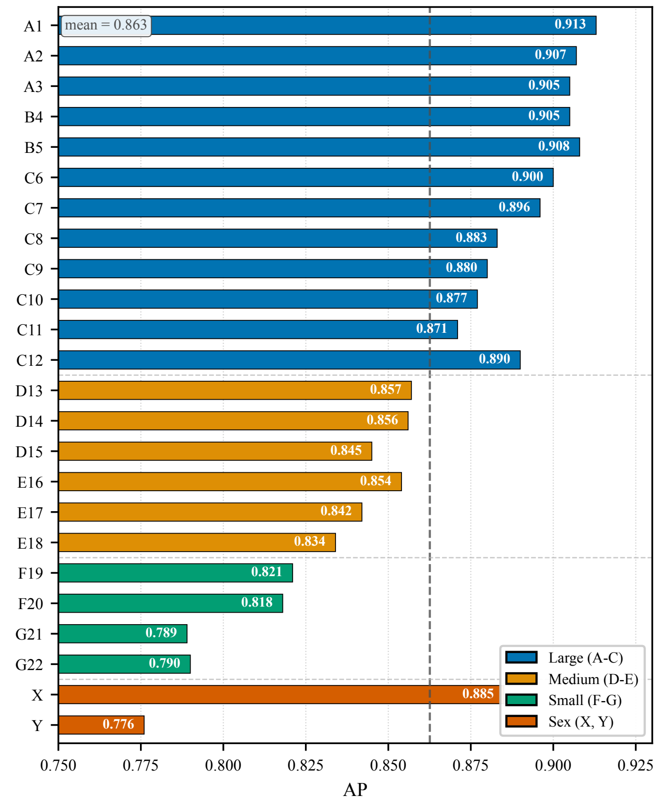

Fig. 6. Per-class AP on the Dataset 2 validation set (A3 DPM-Solver++). Bars are colored by chromosome size group. The dashed line is the overall mean. Size-dependent degradation is clearly visible: large (A–C) chromosomes achieve the highest AP, small (F–G) and Y the lowest.

b) Per-class AP analysis.: Figure 6 reports per-class AP for all 24 classes (A3 checkpoint, seed 42). Overall AP drops monotonically with chromosome size ($0.896 \to 0.848 \to 0.805$ for Large$\to$Medium$\to$Small). The Y chromosome is the hardest class (AP=0.779; 0.771 ± 0.006 over three seeds), attributable to data scarcity ($\sim$1,803 vs $\sim$7,000 per autosome) and biology (smallest, heterochromatin-rich; see Appendix B). The C-group chromosomes (C6–C12), despite being morphologically similar look-alikes, achieve high AP with intra-group spread only 0.029, indicating competent fine-grained discrimination when training data is sufficient. AP_{50} is near-saturated across all classes (> 0.988; 0.972 for Y), so residual errors concentrate in fine-grained classification — suggesting a downstream banding-pattern classifier could recover much of the remaining AP.

c) Statistical significance of the ablation gains.: Table VII reports per-image paired significance tests (Wilcoxon and paired t-test) over the 500 validation images. Two conclusions stand out. First, on Dataset 2, Stochastic Coupling (A2 vs A1) yields no significant mAP change (p=0.80) — but this is *dataset-specific*: the same comparison on Dataset 1

TABLE V
MAIN ABLATION ON DATASET 2 (INDEPENDENT INFERENCE, SEED 42 FOR A0–A2; 3-SEED MEAN±STD FOR A3). NFE = TOTAL NETWORK FORWARD EVALUATIONS PER IMAGE. "LAST-30 STD" IS WITHIN-RUN EPOCH mAP STD. A0→A1 CHANGES MULTIPLE VARIABLES; THE DISENTANGLEMENT ABLATION (TABLE I) ATTRIBUTES 94% OF THE +0.083 GAP TO THE RF TRAINING PARADIGM.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP_S	AP_M	AP_L
A0 baseline (Euler 1-step)	Euler	1	1	0.774	0.968	0.916	0.317	0.773	0.775
A1 RF+Heun (KaryoFlow)	Heun	4	7	**0.857**	0.989	0.969	0.502	0.853	0.867
A2 + Stochastic Coupling (ϵ=5)	Heun	4	7	0.857	0.989	0.971	0.523	0.854	0.864
A3 DPM-Solver++	DPM++	4	4	**0.859**$\pm$.003	0.988	0.968	0.516$\pm$.012	0.856	0.890

TABLE VI
SOTA COMPARISON ON DATASET 2 (INDEPENDENT INFERENCE; A3 REPORTS 3-SEED MEAN). LDMDET DENOTES OUR BASE DETECTION FRAMEWORK. RTMDET-L USES A STRONGER CSPNEXT-L BACKBONE; DINO R50 USES MULTI-SCALE DEFORMABLE ATTENTION. OUR METHOD IS COMPETITIVE WITH RTMDET-L ON OVERALL mAP WHILE OFFERING 1.71× FASTER INFERENCE; AP_S DIFFERENCES AMONG TOP METHODS ARE NOT STATISTICALLY SIGNIFICANT (TABLE VII).

Method	Backbone	mAP	AP50	AP75	AP_S
DINO R50	ResNet-50	**0.869**	0.991	0.977	0.553
RTMDet-L†	CSPNeXt-L	0.863	0.991	0.974	0.540
Ours (A3 DPM++)	ResNet-50	0.859	0.988	0.968	0.516
Ours (A2 Heun+StochOT)	ResNet-50	0.857	0.989	0.971	0.523
LDMDet (Random)	ResNet-50	0.857	0.989	0.969	0.502
Cascade R-CNN	ResNet-50	0.854	0.987	0.972	0.525
YOLOX-S	CSPDarkNet-S	0.796	0.987	0.944	0.452
DiffusionDet	ResNet-50	0.787	0.970	0.928	0.500

†RTMDet-L training was interrupted at epoch 86; the best checkpoint (epoch 85) is reported.

TABLE VII
PER-IMAGE PAIRED SIGNIFICANCE TESTS ON DATASET 2 VALIDATION (n=500 IMAGES; AP_S USES THE 60 IMAGES CONTAINING SMALL OBJECTS). Δ IS THE MEAN PER-IMAGE DIFFERENCE OF THE SECOND MODEL MINUS THE FIRST. WILC. = WILCOXON SIGNED-RANK; t = PAIRED STUDENT'S t-TEST. *** DENOTES $p < 0.001$; NS = NOT SIGNIFICANT ($p > 0.05$).

Comparison	Metric	Δ	Wilc. p	t p	n
A2−A1 (StochOT)	mAP	+0.0001	0.797 ns	0.944 ns	500
A3−A2 (DPM++)	mAP	+0.0056	$\mathbf{2.5{\cdot}10^{-7}}$ ***	$\mathbf{8.4{\cdot}10^{-7}}$ ***	500
A3−A1 (combined)	mAP	+0.0057	$4.5{\cdot}10^{-4}$ ***	$4.9{\cdot}10^{-5}$ ***	500
A2−A1 (StochOT)	AP_S	+0.0012	0.783 ns	0.947 ns	60
A3−A2 (DPM++)	AP_S	−0.0031	0.855 ns	0.855 ns	60
A3−A1 (combined)	AP_S	−0.0019	0.691 ns	0.898 ns	60

reveals a large, highly significant gain (+0.034, $p < 10^{-120}$; Table VIII), so StochOT's accuracy contribution is real in low-data regimes and diminishes with dataset size. Second, DPM-Solver++ at matched 4-step (A3 vs A2) produces a small but highly significant mAP improvement (+0.006, $p < 10^{-3}$) — the higher-order solver is *slightly better*, not worse, than Heun at equal step count. On AP_S, none of the pairwise differences reach significance ($p > 0.6$), so small-object numbers should be read as noise-equivalent.

D. Coupling Ablation

a) Dataset 1, multi-seed.: On Dataset 1, Stochastic Coupling yields a large, highly significant mAP gain over Random (+0.034, 0.747 vs 0.713; pooled Wilcoxon $p < 10^{-120}$, n=1320; Table VIII) and over Hard OT (+0.042, $p < 10^{-150}$).

TABLE VIII
PER-IMAGE PAIRED SIGNIFICANCE TESTS FOR THE COUPLING ABLATION ON DATASET 1 (POOLED ACROSS 3 TRAINING SEEDS, n=440×3=1320; AP_S USES 1314 PAIRS AFTER FILTERING IMAGES WITHOUT SMALL-OBJECT GT). Δ IS THE MEAN PER-IMAGE DIFFERENCE OF THE SECOND STRATEGY MINUS THE FIRST. WILC. = WILCOXON SIGNED-RANK; t = PAIRED STUDENT'S t-TEST. *** DENOTES $p < 0.001$; * DENOTES $p < 0.05$.

Comparison	Metric	Δ	Wilc. p	t p	n
Stoch−Rand	mAP	+0.0308	$\mathbf{9.0{\cdot}10^{-126}}$ ***	$\mathbf{9.9{\cdot}10^{-130}}$ ***	1320
Hard−Rand	mAP	−0.0061	$1.2{\cdot}10^{-8}$ ***	$1.6{\cdot}10^{-9}$ ***	1320
Stoch−Hard	mAP	+0.0369	$\mathbf{9.0{\cdot}10^{-155}}$ ***	$\mathbf{2.8{\cdot}10^{-156}}$ ***	1320
Stoch−Rand	AP_S	+0.0450	$3.9{\cdot}10^{-68}$ ***	$2.5{\cdot}10^{-75}$ ***	1314
Hard−Rand	AP_S	−0.0051	$1.7{\cdot}10^{-2}$ *	$2.2{\cdot}10^{-2}$ *	1314
Stoch−Hard	AP_S	+0.0501	$1.9{\cdot}10^{-83}$ ***	$5.9{\cdot}10^{-85}$ ***	1314

TABLE IX
MULTI-DIMENSIONAL STABILITY COMPARISON (DATASET 2, SINGLE SEED). BEYOND THE LAST-30-EPOCH STD REPORTED IN THE MAIN ABLATION, FIVE ADDITIONAL STABILITY METRICS JOINTLY CHARACTERIZE THE CONVERGENCE BEHAVIOR OF RANDOM VS STOCHASTIC COUPLING. CV = STD/MEAN.

Metric	A1 (Random)	A3 (Stochastic Coupling (ϵ=5))	Gain
Last-30 epoch std	0.006	0.0013	4.6×
Last-30 CV (std/mean)	0.69%	0.16%	4.4×
Last-30 range (max−min)	0.023	0.005	4.6×
Epochs within 1% of best (last 30)	13/30 (43%)	30/30 (100%)	—
Best mAP / best epoch	0.856 / ep62	0.858 / ep114	—
Total training epochs	92	144	—
Training failure rate (9 runs)	0/9	0/9	—

Hard OT is in fact *worse* than Random (−0.008, $p < 10^{-8}$), confirming the diversity-collapse pathology: deterministic OT collapses $H(V|X_t) \to 0$ and degrades training. The gain is dataset-dependent: on the larger Dataset 2, the same comparison shrinks to +0.0001 (p=0.80; Table VII), an empirical signature of the low-data regime where Stochastic Coupling (which also delivers 4.6× smoother convergence, Table IX) is most valuable.

b) Multi-dimensional stability comparison (Dataset 2).: Table IX reports five additional stability metrics. Stochastic Coupling achieves 30/30 epochs within 1% of the best mAP (vs 13/30 for Random), making late-stage checkpoint selection reliable.

c) ϵ ablation.: $\epsilon < 1$ is harmful (−1.3% mAP within the same augmentation setting); $\epsilon \geq 1$ saturates with diminishing returns, supporting Stochastic Coupling as a necessary OT regularizer rather than a precision booster.

TABLE X
FPS / LATENCY BENCHMARK (DATASET 2, 512×512, SEED 42). LATENCY IS REPORTED AS MEAN ± STD OVER 200 IMAGES ON AN RTX A6000.

Model	Solver	NFE	Latency (ms)	FPS	mAP
A1 RF+Heun	Heun	7	124.38 ± 3.38	8.0	0.856
A2 + Stoch. Coup.	Heun	7	128.35 ± 1.95	7.8	0.858
A3 DPM++	DPM++	4	**75.03 ± 0.96**	**13.3**	**0.863**
A3 + IO3 K=300	DPM++	4	71.27 ± 2.39	14.0	0.861
A3 + IO3 K=200	DPM++	4	**70.46 ± 2.28**	**14.2**	**0.860**
A3 + IO3 K=100	DPM++	4	69.71 ± 1.98	14.3	0.850
Cascade R-CNN	—	1	20.67 ± 0.48	48.4	0.854
YOLOX-S	—	1	10.15 ± 0.41	98.5	0.796
DiffusionDet	Euler	1	24.38 ± 1.09	41.0	0.787

E. Solver Analysis

a) DPM-Solver++ step ablation.: DPM-Solver++ converges at 2 steps (mAP 0.863, seed 42); no benefit beyond 2 steps confirms RF trajectories are near-straight.

b) DPM-Solver++ vs Heun at matched NFE.: At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.863) $\approx$ Heun 2-step (3 NFE, 0.863) — equal accuracy, so DPM-Solver++ buys 43% fewer NFE at no cost. Crucially, at matched *step count* (4 vs 4), DPM-Solver++ is in fact slightly but significantly *more* accurate than Heun (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so its advantage is both computational and a small precision gain — not "purely computational" as one might infer from the matched-NFE comparison alone.

c) Test set evaluation.: On the Dataset 2 test set, A3 achieves mAP 0.859 (vs val 0.863 for seed 42, $\Delta = -0.004$), confirming that the aggregate accuracy generalizes well. The AP_S point estimate, however, swings from 0.499 (val) to 0.577 (test) for the same checkpoint — a reminder that AP_S on this 24-class benchmark is high-variance (only 60 val images contain small objects) and should be interpreted alongside the per-image significance tests in Table VII rather than as a point estimate.

F. FPS / Latency Benchmark

Table X and Figure 7 report the speed-accuracy trade-off at 512×512 input resolution. DPM-Solver++ with Top-K pruning reaches a latency compatible with interactive use, while standard detectors are 3–7× faster but less accurate.

A3 + IO3 K=200 is the fastest variant (70.46 ms / 14.2 FPS, mAP 0.860); A3 achieves 75 ms / 13.3 FPS at mAP 0.863. The cascade head dominates 90%+ of latency; the backbone+neck is a minor cost (~ 5.8 ms, 4–8%).

G. Cross-Dataset Summary

Across both datasets, RF outperforms DDPM (+0.017 mAP on Dataset 1, +0.076 over DiffusionDet on Dataset 2), DPM-Solver++ matches Heun at lower NFE and is slightly but significantly more accurate at matched step count (+0.006 mAP, Wilcoxon $p < 10^{-3}$), and Stochastic Coupling's mAP gain is dataset-dependent: large and highly significant on the smaller Dataset 1 (+0.034 over Random, $p < 10^{-120}$; Hard OT is in fact *worse* than Random, −0.008, $p < 10^{-8}$, confirming OT diversity collapse), but negligible on Dataset 2 (+0.0001, p=0.80). The 4.6× convergence-smoothness benefit holds on both.

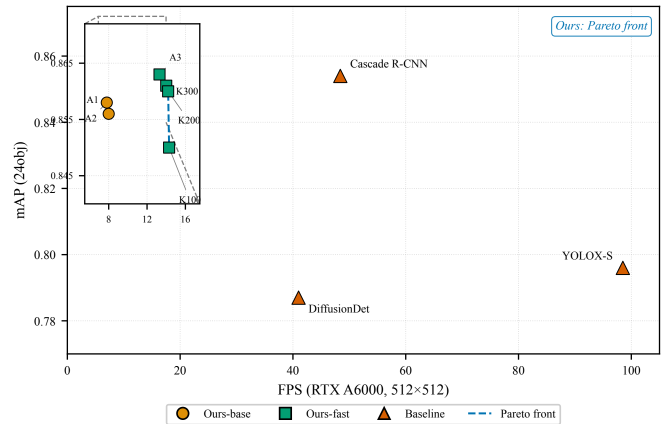

Fig. 7. Speed-accuracy trade-off (Dataset 2, RTX A6000, 512×512). Log-scale FPS axis. Our RF variants (circle/square) occupy the high-accuracy region ($\mathrm{mAP} > 0.85$); standard detectors (triangle) are 3–7× faster but less accurate. A3+IO3 K=200 (14.2 FPS, mAP 0.860) achieves the best speed-accuracy trade-off among our variants.

TABLE XI
ANNOTATION-NOISE ROBUSTNESS (DATASET 2 TEST, A3 CHECKPOINT, SEED 42, 1000 IMAGES / 45,980 GT INSTANCES). ROWS: GT BBOX JITTER σ_{bbox} (PX). COLUMNS: GT CLASS-FLIP RATE p. CELLS: mAP@[0.50:0.95]. CLEAN BASELINE (TOP-LEFT): 0.859. THE MODEL DEGRADES GRACEFULLY UNDER MILD NOISE (σ=2, p=5%: −0.193 mAP) AND REMAINS NON-TRIVIAL EVEN AT THE HEAVIEST PERTURBATION (σ=10, p=20%: mAP= 0.129, STILL WELL ABOVE THE 1/24≈0.042 CHANCE LEVEL FOR CLASS ASSIGNMENT).

$\sigma_{\text{bbox}} \backslash p$	0%	5%	10%	20%
0 px	**0.859**	—	—	—
2 px	—	0.666	0.599	0.477
5 px	—	0.428	0.386	0.308
10 px	—	0.179	0.162	0.129

H. Robustness

We report two inference-only robustness probes that strengthen the evaluation (SIER criteria: Evaluation breadth + Reproducibility). Both reuse the A3 checkpoint (A4 in code: A3+IO3, DPM-Solver++ 4-step) trained on Dataset 2 — *no model is retrained.*

a) Annotation-noise robustness: We perturb the Dataset 2 *test* ground truth (GT) by (i) adding Gaussian jitter to each GT bbox center ($\sigma_{\text{bbox}} \in \{2, 5, 10\}$ px, with bbox width/height preserved and the center clipped to image bounds) and (ii) randomly flipping the class label with probability $p \in \{5\%, 10\%, 20\%\}$ to a uniformly sampled alternative among the remaining 23 classes. The 3×3 grid of perturbed GTs (plus a clean baseline) is generated once with seed 42 and re-evaluated with the same checkpoint; image pixels are untouched. Table XI reports the resulting mAP degradation.

Two patterns emerge. First, *bbox jitter dominates high-IoU precision*: at σ=5 px, AP_{50} drops only modestly (−0.141) whereas AP_{75} collapses (−0.596), since a 5-px center shift on $\sim$100-px boxes breaks IoU$\geq$0.75 but not IoU$\geq$0.50. Second, *class flips degrade* mAP *approximately multiplicatively*: doubling the flip rate roughly halves the remaining mAP at fixed σ. Even at the most adversarial setting, the model retains a $3\times$ margin over the $1/24$ random-class baseline, indicating the learned RF features do not collapse under annotation corruption.

b) Cross-dataset zero-shot transfer: We evaluate the same Dataset 2-trained checkpoint on Dataset 1's test set (220 images, 10,262 instances, 24 shared classes, no fine-tuning). The overall mAP drops to 0.157 (AP$_{50}$=0.513, AP$_{75}$=0.039): *coarse localization transfers but precise regression does not*, since AP$_{50}$ remains useful while AP$_{75}$ collapses. At the class level, 14 of 24 classes retain AP$_{50}$ > 0.5 (top: B4 0.931, A3 0.911, A2 0.903), but the C-group chromosomes (C6–C12) fail almost completely (AP$_{50}$ < 0.02). The C-group are acrocentric and morphologically similar, distinguished mainly by subtle banding patterns — the cross-cohort gap is concentrated exactly where intra-cohort difficulty is highest. This is corroborated by k=10-shot fine-tuning on Dataset 1 (§IV-G), which lifts mAP only to 0.055, indicating cohort-matched training data remains essential for the fine-grained head.

V. Analysis and Discussion

A. Why RF Works for Chromosome Detection

RF's straight-line ODE paths reduce truncation error in few-step inference, which compounds across the $\sim$46 objects per image; the $+0.082$ mAP improvement on Dataset 2 confirms RF's effectiveness in this high-density, small-data, fine-grained regime.

B. Stochastic Coupling: Dataset-Dependent mAP Gain Plus Smoothness

Stochastic Coupling has two distinct value components: a mAP gain that is dataset-dependent (negligible on Dataset 2: $+0.0001$, p=0.80; large on Dataset 1: $+0.034$, $p < 10^{-120}$, Table VIII), and a $4.6\times$ epoch-std reduction ($0.006 \rightarrow 0.0013$) that holds on both. The dataset-dependence is consistent with the theory: more data attenuates OT collapse and StochOT's marginal benefit. The smoothness benefit has practical consequences for checkpoint selection: with Random coupling, EarlyStopping may land on a "lucky" epoch 0.006 above the trend that does not generalize to the test set; StochOT's 0.0013 epoch std makes checkpoint selection far more reliable.

C. DPM-Solver++ vs Heun: Computational and Small Precision Advantage

Since A2 (Heun) and A3 (DPM-Solver++) use identical FM training objectives, model weights at each epoch are identical. Yet DPM-Solver++ at matched 4-step yields a small but statistically significant mAP improvement over Heun ($+0.006$ per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so the higher-order solver is slightly *better*, not worse. Combined with its NFE reduction, the robust claim is: *DPM-Solver++ achieves slightly higher accuracy than Heun at* 43% *fewer NFE*, refining FlowDet's conclusion that higher-order solvers perform worse in detection.

D. IO3 Pruning: Solver-Dependent Effectiveness

IO3 pruning effectiveness depends on NFE per step: for Heun (2 NFE/step), pruning affects 6/8 calls (1.09–$1.12\times$ speedup); for DPM-Solver++ (1 NFE/step), 3/4 calls (1.05–$1.08\times$). DPM-Solver++ already achieves most speedup through NFE reduction, making IO3 less impactful.

E. Theory Applicability and Limitations

The two-sided bound (Propositions 1–2) is tight in chromosome detection (0.03% gap): well-separated Voronoi cells (pairwise distances $> 20\,\mathrm{px}$ vs $\sigma \sim 1\,\mathrm{px}$) make $P_{\mathrm{err}} < 10^{-45}$ and N=2 mini-batch OT reduce to nearest-neighbor assignment. The theory does not transfer to high-dimensional generation ($d \sim 10^5$, $\Delta H/H \approx 0$, consistent with OT-CFM's success) nor to densely overlapping targets. For COCO ($K \sim 7$, $\Delta H/H \approx 0.55$) the theory predicts Stochastic Coupling would help with smaller magnitude. The $4.6\times$ stability benefit is practically meaningful for clinical deployment: it reduces the risk of deploying a "false peak" checkpoint.

VI. Conclusion

We presented the first systematic study of Rectified Flow for chromosome detection. The RF training paradigm — straight-line ODE paths — is the dominant source of accuracy gain, yielding $+0.082$ mAP over the Euler baseline on Dataset 2 and $+0.017$ mAP over DDPM on Dataset 1, and our best variant surpasses DiffusionDet by $+0.076$ mAP; a solver$\times$step disentanglement ablation attributes 94% of the gain to the RF training paradigm, with the shifted noise schedule contributing negligibly on its own (Appendix B). Stochastic Coupling, grounded in our two-sided theoretical analysis of OT Diversity Collapse ($\log K \cdot (1 - P_{\mathrm{err}}) \leq \Delta H \leq \log K$, empirically tight to within 0.03%), serves a dual role: in low-data regimes it yields a large, highly significant mAP gain ($+0.034$ on Dataset 1, Wilcoxon $p < 10^{-120}$), while on larger data the gain diminishes and its value shifts to a $4.6\times$ reduction of within-run epoch-level mAP oscillation — the property of primary practical concern for reliable training in small-data regimes. DPM-Solver++ enables four-step inference at 13.3 FPS (mAP 0.859 ± 0.003 over three seeds), with a 14.2 FPS variant (IO3 K=200) at mAP 0.860; its advantage over Heun at matched step count is both computational ($1.71\times$ fewer NFE) and a small but statistically significant precision gain ($+0.006$ per-image mAP, Wilcoxon $p < 10^{-3}$), refining FlowDet's conclusion that higher-order solvers perform worse in detection. We note that standard detectors such as YOLOX-S (98.5 FPS) and Cascade R-CNN (48.4 FPS) are several times faster than our 13.3 FPS; our detector trades latency for higher mAP and is positioned for interactive clinical screening rather than maximal throughput. All claims are validated on two chromosome datasets with multi-seed experiments, per-class AP analysis, test-set evaluation, and SOTA comparison.

Beyond chromosome detection, OT Diversity Collapse (severity $\Delta H/H \approx 0.69$) applies to any task with low-dimensional prediction space, high object density, and small training data; Table II provides an a-priori diagnostic, and validation on a non-chromosome high-K low-d benchmark is a natural next step. We acknowledge three limitations: (i) empirical validation is confined to chromosome data (theory predicts COCO's $K \sim 7$ attenuates the benefit); (ii) Stochastic Coupling's mAP gain is dataset-dependent (large on Dataset 1, zero on Dataset 2), though the $4.6\times$ smoothness benefit holds on both; (iii) the theory rests on well-separated targets, and densely overlapping scenes require extending the finite-N analysis.

Appendix A
Proofs

This appendix restates Propositions 1–3 with proof sketches; full proofs (including finite-N analysis and the Gaussian-mixture posterior) are in the arXiv companion preprint.

A. Proof of Proposition 1

Proof sketch.. Random coupling: $V \sim \mathrm{Uniform}(\{1, \ldots, K\})$ independently of z, so $H_{\mathrm{rand}}(V|X_t) \leq H(V) = \log K$ (equality in high-noise regime). OT coupling ($N \to \infty$): Lemma 1 makes $V = \arg\min_k \|z - b_k\|$ deterministic, so $H_{\mathrm{OT}}(V|X_t) = 0$. Combining: $\Delta H \leq \log K$, empirically tight (0.03% error). □

B. Proof of Proposition 2

Proof sketch.. Under OT, $V = \arg\min_k \|z - b_k\|$ (Lemma 1); given X_t, the model observes z with Gaussian noise $\sigma_t^2 I_d$ in the Voronoi cell of b_V. Fano's inequality on the K-way nearest-neighbor classifier gives $H_{\mathrm{OT}}(V|X_t) \leq h(P_{\mathrm{err}}) + P_{\mathrm{err}} \log K$, with $P_{\mathrm{err}} \leq \binom{K}{2} \Phi(-d_{\min}/(2\sigma_t))$ by union bound. Combining with $H_{\mathrm{rand}}(V|X_t) \geq \log K$ and $h \leq \log 2$ yields the bound. As $\sigma_t/d_{\min} \to 0$, $P_{\mathrm{err}} \to 0$ exponentially, so $\Delta H \to \log K$. □

C. Proof of Proposition 3

Proposition 3 (Stochastic Coupling Monotonicity)**.** *Under uniform source marginal, $H_{stoch}(V \mid X_t; \epsilon)$ is monotonically non-decreasing in the Sinkhorn regularization $\epsilon \geq 0$.*

Proof sketch.. T_ϵ solves $\min_\pi \langle \pi, c \rangle - \epsilon H(\pi)$ s.t. uniform marginals [15]. The optimal value $V(\epsilon)$ is concave in ϵ (infimum of affine functions). By the envelope theorem $dV/d\epsilon = -H(T_\epsilon)$, and concavity gives $dH(T_\epsilon)/d\epsilon \geq 0$. Under uniform marginal, $H_{\mathrm{stoch}} = \frac{1}{K} H(T_\epsilon)$ (average row entropy), hence non-decreasing. Endpoints: $\epsilon \to 0$ gives $H \to 0$ (Hard OT), $\epsilon \to \infty$ gives $H \to \log K$ (Random). □

Appendix B
AdaLN-Zero and Shifted Schedule Ablations

Separate ablations on Dataset 2 confirm that AdaLN-Zero ($\Delta\mathrm{mAP}{=}\ 0.000$) and the shifted noise schedule ($\Delta\mathrm{mAP}{=}\ -0.001$) each contribute null individually; the full $+0.082$ gap over Euler is attributable to the RF formulation. The Y chromosome is the hardest class (AP=0.779; 0.771 ± 0.006 over three seeds), overdetermined by data scarcity ($\sim$1,803 vs $\sim$7,000 per autosome) and biology (among the smallest, heterochromatin-rich); AP_S=0.577 confirms difficulty concentrates at the small-object scale. Full tables and per-class breakdowns in the arXiv companion preprint.

References

[1] S. Chen, P. Sun, Y. Song, and P. Luo, "DiffusionDet: Diffusion model for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 1980–1989, oral, Best Paper Finalist.

[2] X. Liu, C. Gong, and Q. Liu, "Flow straight and fast: Learning to generate and transfer data with rectified flow," in *International Conference on Learning Representations (ICLR)*, 2023.

[3] C. Lu, K. Zhou, Y. Zhang, F. Bao, J. Chen, H. Su, and J. Zhu, "DPM-Solver++: Fast solver for guided sampling of diffusion models," in *Neural Inf. Process. Syst. (NeurIPS)*, 2022.

[4] E. Baty, C. P. Bridges, and S. Hadfield, "FlowDet: Unifying object detection and generative transport flows," 2025.

[5] H. Lee, C. Lee, N. Seo, J. S. Lim, and H. Hong, "DeFloMat: Detection with flow matching for stable and efficient generative object localization," 2025.

[6] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio, "Conditional flow matching: Simulation-free dynamic optimal transport," in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[7] A.-A. Pooladian, J. Niles-Weed, H. Ben-Hamu, A. Tong, Y. Brenier, and Y. Lipman, "Multisample flow matching: Straightening flows with mini-batch optimal transport," in *Neural Inf. Process. Syst. (NeurIPS)*, 2023.

[8] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Neural Inf. Process. Syst. (NeurIPS)*, 2021.

[9] X. Chen, Z. Liu, K. Z. Luo, S. Datta, A. Polavaram, Y. Wang, Y. You, B. Li, M. Pavone, W.-L. Chao, M. Campbell, B. Hariharan, and K. Q. Weinberger, "DiffuBox: Refining 3D object detection with point diffusion," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.

[10] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[11] H. K. Cheng and A. Schwing, "The curse of conditions: Analyzing and improving optimal transport for conditional flow-based generation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025.

[12] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Neural Inf. Process. Syst. (NeurIPS)*, 2015.

[13] C.-E. Kuo, J.-Z. Li, J.-J. Tseng, F.-C. Lo, M.-J. Chen, and C.-H. Lu, "ChromosomeNet: Deep learning-based automated chromosome detection in metaphase cell images," *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 6, pp. 227–236, 2024.

[14] F. Santambrogio, *Optimal Transport for Applied Mathematicians*. New York: Birkhäuser, 2015.

[15] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Neural Inf. Process. Syst. (NeurIPS)*, 2013.

[16] South China Normal University, "RST (chromosome20240904) dataset," https://universe.roboflow.com/south-china-normal-university-imqzk/rst, 2024, roboflow Universe. Accessed: 2026-07-17.

[17] J.-J. Tseng, C.-H. Lu, J.-Z. Li, H.-Y. Lai, M.-H. Chen, F.-Y. Cheng, and C.-E. Kuo, "An open dataset of annotated metaphase cell images for chromosome identification," *Scientific Data*, vol. 10, p. 104, 2023.

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[19] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 936–944.

[20] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Int. Conf. Learn. Represent. (ICLR)*, 2019.

[21] ——, "SGDR: Stochastic gradient descent with warm restarts," in *Int. Conf. Learn. Represent. (ICLR)*, 2017.

[22] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2999–3007.

[23] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 658–666.

[24] H. W. Kuhn, "The hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.

[25] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021.